



of 1 TOPS@INT8 (1 TOPS = 1 trillion operations per second), 256MB DDR3 RAM, TFT touch screen, SD card, microphone, 5MP camera, and 2.4GHz Wi-Fi.

The board runs a lightweight, tailor-made Linux-based operating system for Kendryte K210 boards. Key features include integrated support for YOLOv8, headless mode for efficient remote operations, Python 3.11 for scripting and automation, and an SSH daemon for secure remote access. This setup is ideal for real-time face recognition, making it suitable for entry-level AI solutions or lightweight edge AI systems.

YOLO and other modules typically require manual setup, but in MaixCam, YOLO is already installed within its operating system.

Software

The system utilises two pre-trained MaixCam models: yolov5s.mud (which detects general objects, including humans without a hard hat) and yolov5s_hat.mud (which detects hard hats). By combining these two models, individuals are classified into two categories: 'person with a hard hat' and 'person without a hard hat.'

During operation, MaixCam captures an image of the scene, which is then processed through both models. The system filters and identifies objects based on the two classifications and counts the detected individuals in each category.

Field testing

The device has been tested with a portable power bank, successfully identifying both classifications in real-world conditions. Initial concerns regarding whether such



Fig. 3: Helmet on human head is correctly identified by green box, while on the right, a dog, being non-human, is utterly ignored

a compact device could detect multiple figures in a single frame were addressed through successful trials, confirming the system's capability to handle the task effortlessly.

To test the system, power on the device and allow the camera to capture a head in the frame. The system then compares it with the stored database. If the object is not found, the red LED lights up and it is not counted. However, if the object is recognised, the green LED lights up and it is counted.

Aftermath

In the field of safety behavioural science, research has shown that job supervision is more effective

in preventing accidents than job training. If supervision can be entrusted to machines, it offers a significant advantage: machines operate diligently and tirelessly without requiring rest. This advancement aligns with future workplace trends, where machines are expected to handle repetitive and monotonous tasks, enabling humans to focus on skill-based roles that demand creativity and judgement.

However, one crucial consideration remains. Unlike a human safety supervisor, a machine may not initially command the same level of respect. People may be more inclined to ignore an automated system, whereas a supervisor composed of flesh and blood naturally enforces authority. Overcoming this behavioural gap is essential for ensuring that machines are as effective as human counterparts in enforcing safety compliance. **EFY**

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